



DRAGON'S DOMINANCE



98% of global OSAT capacity is located in the Asia region



Within Asia, it is concentrated mostly in China, with a 47.39% share. Both Taiwan and US-based OSATs have multiple sites in China



China has three OSAT players in next-generation advanced packaging, 11 players in the commercial or current age advanced packaging, and five players in the legacy playing spaces

SOURCE FAB ECONOMICS, GLOBAL SEMICONDUCTOR POLICY COUNCIL

ATMP unit by Micron. The planned HCL-Foxconn OSAT facility in India is worth a third of Micron's planned investment in the ATMP in Gujarat, but the HCL-Foxconn project will have a much bigger market.

188 | Micron ATMP's output is restricted to Micron products as it is an IDM, not a foundry or OSAT, says Danish Faruqui, CEO of Fab Economics, a US-based boutique advisory for semiconductor fab/ATMP/OSAT greenfield projects. "Micron's ATMP will add only to supply chain resiliency for Micron and its customers. But the HCL-Foxconn OSAT will provide assembly, packaging, and testing to a whole array

of customers like a foundry model, thus enhancing supply chain resiliency for multiple players in India and globally," says Faruqui.

Also, while Micron ATMP's output is restricted primarily to memory die type assembly, packaging and testing, HCL-Foxconn's OSAT will offer fungibility for assembly, packaging and testing of logic, memory, analog sensors, and discreet die types. So India will offer the full portfolio diversity to design, manufacture, package and test products for all end markets.

OSATs from Indian players will be large assembly, packaging and testing units that could provide

service to potential fabs in India needing OSAT services. For the design start-ups sprouting with government support, these OSATs can help complete India's manufacturing-assembly-packaging-test value chain domestically. In short, India will sit on a trusted value chain.

CHINA'S SUPREMACY

Fab Economics says 98% of the world's OSAT capacity is physically located in Asia and concentrated in China, whether the companies owning them are headquartered in Taiwan or the US. "It is important to understand the China OSAT play to inform strategy and plan-

IT SUPPORT

The three main areas of service being provided by IT companies in the semiconductor ecosystem

PRE-SILICON DESIGN

- **FRONT END:** SoC design, prototyping and validation
- **BACKEND:** Physical design, verification, synthesis and timing analysis
- **ANALOG & MIXED SIGNAL:** Circuit design and layout

POST-SILICON VALIDATION AND EMBEDDED DEVELOPMENT

- Post-silicon functional validation, automated test engineering, characterisation and certification
- Embedded hardware and software
- OSAT development

AFTERMARKET SUPPORT

- System re-engineering and obsolescence management
- Value engineering and sustenance engineering

SOURCE TECH MAHINDRA